Ivan **Ovinnikov**

APPLIED SCIENTIST · REINFORCEMENT LEARNING SIMULATION & PHYSICAL AI

Zürich, Switzerland

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SUMMARY

Applied scientist working at the intersection of reinforcement learning, generative models, and simulation-based learning. Interested in robust learning dynamics, scalable training systems, and sequential decision-making under distribution shift. Combines theory-driven algorithm design with practical implementation and rigorous empirical evaluation.

EXPERIENCE

ANYbotics AG — Zürich, Switzerland

MACHINE LEARNING ENGINEER (RL FOR LOCOMOTION) — Oct 2024 – Present

- Developed RL locomotion controllers in IsaacSim/IsaacLab for quadrupeds deployed in industrial inspection.
- Built and improved components of a GPU-accelerated training and evaluation workflow enabling fast sim-to-real iteration.
- Led and secured a EuroHPC grant enabling up to 50,000 H100 GPU-hours for large-scale RL robustness and scaling experiments.
- Designed and implemented novel safety curriculum learning techniques to improve stability under rare failure modes.

ETH Zürich — Zürich, Switzerland

DOCTORAL CANDIDATE (DR. SC. IN COMPUTER SCIENCE) — Feb 2019 – Jun 2024

- Proposed and implemented an inverse reinforcement learning framework for surgical trainee evaluation and skill assessment in simulation-based training.
- Developed inverse reinforcement learning methods for learning robust objectives from diverse demonstrations under dynamics and environment shift.
- Studied reward generalization, imitation learning, and Wasserstein/distributional objectives for learning from demonstrations.
- Published work on RL benchmarks, causally invariant reward learning, and imitation learning with sliced Wasserstein distances.
- Teaching Assistant for Advanced ML, Statistical Learning Theory and Algorithmic Game Theory

Disney Research — Zürich, Switzerland

RESEARCH ASSISTANT — Nov 2016 – Dec 2018

- Extended sequence-to-sequence NLP models with structured variational inference for latent structure and uncertainty modeling.
- Investigated hyperbolic geometry for Wasserstein autoencoders to learn hierarchical representations for large-scale NLP.

EDUCATION

ETH Zürich — Zürich, Switzerland

DR. SC. IN COMPUTER SCIENCE — Feb 2019 – Jun 2024

M.SC. IN ELECTRICAL ENGINEERING AND INFORMATION TECHNOLOGY — Sep 2009 – Jun 2015

SKILLS

Research	Reinforcement learning, imitation learning, reward learning, curriculum design, robust learning under distribution shift, generative and representation learning
LLM / Post-training	VLA/world-model evaluation | SFT/RLHF/DPO/RLVR fundamentals | data curation, reward design, and reasoning-action alignment.
Systems & Simulation	PyTorch, JAX, TensorFlow | GPU-accelerated training workflows | experiment tracking / orchestration | IsaacLab/IsaacSim, MuJoCo, ROS, Unity ML-Agents
Programming	Python, C++, C#, Java, C, MATLAB
Languages	Russian (native) | English (C2) | German (C2) | French (C1) | Italian (B2)

SELECTED RESEARCH THEMES

- **Open-ended learning:** automated curricula over task distributions for long-tail robustness and skill discovery.
- **Distributional control:** generative and control-as-inference approaches to scalable sequential decision-making.
- **Robust learning signals:** reward, preference, and evaluation design under distribution shift.

SELECTED PUBLICATIONS

I. Ovinnikov, A. Beuret, F. Cavaliere, and J. M. Buhmann (2024). "Fundamentals of Arthroscopic Surgery Training and beyond: a reinforcement learning exploration and benchmark". In: International Journal of Computer Assisted Radiology and Surgery.

I. Ovinnikov, E. Bykovets, and J. M. Buhmann (2024). "Learning Causally Invariant Reward Functions from Diverse Demonstrations". In: arXiv preprint arXiv:2409.08012.

I. Ovinnikov, A. Terenin, and J. M. Buhmann (2024). "Imitation Learning using Generalized Sliced Wasserstein Distances". In: TMLR, In review.

I. Ovinnikov (2019). "Poincare Wasserstein Autoencoder". In: arXiv preprint arXiv:1901.01427.